

Kartik Joshi

linkedin.com/in/kartikjoshi23 | github.com/KartikJoshi23 | kartik-joshi.vercel.app
+971-565582009 | kartujoshi2002@gmail.com | Dubai, UAE

Summary

AI Engineer spanning graph deep learning, reinforcement learning, real-time systems, and full-stack deployment – most recently building an end-to-end graph neural network platform for financial and procurement crime detection. Smart India Hackathon Finalist, ICC NIUM Global Top 10, and IJANA-published ML researcher. Completing a Masters in AI with Business at SP Jain Dubai, targeting applied AI engineering roles in the UAE market.

Education

SP Jain School of Global Management Dubai, UAE
Masters in AI with Business – 2-Year Postgraduate Programme, CPI: 3.6/4.0 2025 – 2027

Parul University Vadodara, India
B.Tech in Computer Science & Engineering (AI), CGPA: 8.04/10 2020 – 2024

– Selected as University Representative for the National Entrepreneurship Challenge at IIT Bombay.

Experience

Lead AI Engineer Apr 2024 – July 2026
VaultxLedger Technology Solutions Vadodara, India

- Designed and demonstrated a real-time computer vision pipeline for Vadodara Rural Police – a working ANPR prototype built in PyTorch and OpenCV achieving sub-100ms inference latency; received a Certificate of Appreciation from IPS Rohan Anand for the prototype demonstration.
- Directed a team of 6 engineers through computer vision pipeline architecture, hardware integration, and direct coordination with IPS-ranked law enforcement officers – delivering the prototype on schedule without external QA support.
- Converted the engagement into VaultxLedger’s first government client relationship, translating a hackathon prototype into a confirmed pipeline through the PIERC incubation program.

Incubation Program Associate Dec 2023 – Mar 2024
PIERC Startup Incubator Vadodara, India

- Developed go-to-market strategy and investor pitch for VaultxLedger during PIERC Cohort-2; secured incubation support and progressed to a confirmed government client engagement with Vadodara Rural Police.

Projects

CollusionGraph – Graph Deep Learning for Financial & Procurement Collusion Detection 2026
PyTorch, PyTorch Geometric (GATv2, GraphSAGE, R-GCN), PyGOD, FastAPI, React, TypeScript, AWS, NVIDIA NIM

- Built an end-to-end graph deep learning system unifying six public datasets across financial (money laundering) and procurement (bid-rigging) domains into one domain-agnostic graph schema, screening for coordinated networks that no single-record model can detect.
- Combined supervised GNNs (GATv2, GraphSAGE, R-GCN), unsupervised graph anomaly detectors (DOMINANT, GAE), and classical baselines into an isotonic-calibrated ensemble – recovering 92% of planted bid-rigging cartels (vs. 12% for a non-learning baseline) on a 488K-node, 1.45M-edge unlabeled network where classical models could not run at all.
- Shipped as a full product: an explainability layer (PGExplainer + 9 FATF/OECD motif patterns), a tool-grounded GenAI Investigator Copilot passing a 24-question golden-test gate, a React/TypeScript console, and a FastAPI read-only serving layer enforcing a strict train/serve trust boundary.

AlgoViz – Real-Time Algorithmic Trading Dashboard 2025
Python, Streamlit, Plotly, NumPy, SciPy, scikit-learn, Docker, Binance WebSocket API

- Engineered a live trading dashboard streaming BTC/USDT data via the Binance WebSocket API at 50+ trades/sec, computing 9 market microstructure features (VWAP, TWAP, order book imbalance, price velocity, rolling volatility) for signal generation.
- Designed a custom LSTM-Attention network implemented purely in NumPy – zero external deep learning framework dependencies – achieving sub-100ms inference on streaming tick data.
- Delivered a 7-component ensemble predictor with a full P&L backtesting engine, Dockerized deployment, and 45 pytest unit tests.

Smart City Traffic Management – Multi-Agent Reinforcement Learning

2026

Python, React 18, Vite, NumPy, Pandas, TOPSIS

- Designed a Multi-Agent Reinforcement Learning (MARL) system for adaptive traffic signal control across a 16-intersection Dubai grid, training decentralized Q-Learning and SARSA agents over 60,000 steps with a 6-factor reward function balancing queue reduction, throughput, and emissions.
- Built a custom simulation environment with a 3-protocol coordination layer (Green Wave, Conflict Resolution, Corridor Balancing) and a Chaos Mode engine stress-testing sandstorm conditions, corrupted sensors, and emergency vehicle preemption.
- Benchmarked agent policies via WSM and TOPSIS multi-criteria frameworks across 4 weighted metrics; delivered results through a React 18 dashboard with live grid-state animations and Q-value Explainable AI (XAI) panels.

Smart Contract Security Scanner

2026

Python, FastAPI, React 18, TypeScript, Gemini 2.5 Flash, Docker

- Built a full-stack smart contract security scanner for Dubai's DIFC Web3 ecosystem, integrating a React 18 frontend, a FastAPI REST backend, and a Google Gemini 2.5 Flash LLM inference pipeline.
- Implemented a 0–100 risk scoring engine with line-level Solidity vulnerability mapping across 10+ categories (reentrancy, overflow, access control) and automated fix generation across 6 blockchain networks via the Alchemy API.

Publications

AI Cloud Service Recommendation System – IJANA Journal, ASICS Conference

2024

- Co-designed the ML pipeline and model evaluation framework for a cloud service recommendation system, achieving 70%+ classification accuracy across 6 user-defined parameters on a proprietary dataset of 1,000+ records.

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, C/C++, SQL, HTML/CSS
- **AI & Machine Learning:** PyTorch, TensorFlow, scikit-learn, OpenCV, MediaPipe, NumPy, Pandas, SciPy, Reinforcement Learning, LSTM & Attention, Deep SVDD, Explainable AI (XAI)
- **LLM & Generative AI:** Google Gemini API, LLM Inference Pipelines, Prompt Engineering, Responsible AI, AI Chatbot Development
- **Web & DevOps:** React, Vite, FastAPI, SQLAlchemy, Docker, Git, GitHub Actions, REST APIs, Jupyter Notebooks
- **Developer Tools:** Cursor, GitHub Copilot, Claude (Anthropic), Git/GitHub
- **Domains:** Real-Time Systems, Edge Deployment, MLOps, Graph Machine Learning, Web3 & Blockchain, Smart City Systems, Data Analytics

Achievements

- Global Top 10 – ICC NIUM Hackathon, Web3 & Blockchain (International)
- Finalist – Smart India Hackathon 2023 (National, 1M+ participants)
- Winner – Let's Hack India Tour, Parul University

Leadership & Activities

- **AI Presenter & Product Demonstrator**, Hackathons, Incubators & Law Enforcement Engagements, India (2023 – Present): Delivered live technical demonstrations of AI systems to IPS-ranked law enforcement officers, incubator panels, and international hackathon judges across PIERC, IIT Bombay NEC, Smart India Hackathon, and ICC NIUM.
- **Marketing Lead & Event Head**, CDC FunFest, Parul University, Vadodara, India (2022 – 2023): Managed end-to-end operations and led external communications for a cross-departmental event team.
- **Bal Sanchalak (Youth Leader)**, SMVS Swaminarayan Sanstha, Gujarat, India (2019 – Present): Oversaw mentorship programmes for youth cohorts, developing leadership and facilitation skills.

Languages

English (Professional) • Hindi (Conversational) • Gujarati (Native)